Analucía Vera Maric

averamaric@gmail.com – Manchester, UK
https://www.linkedin.com/in/analucia-vera-maric/

PERSONAL PROFILE

Master's student in Robotics & Automation at Manchester Metropolitan University with a background in Industrial Electronic Engineering and international experience. Former IoT Project Manager at Alhona, a consulting company, leading digitalization initiatives for consulting clients. Combines technical expertise with strong project management skills, a proactive mindset, and a commitment to continuous learning through hands-on work and self-study.

LANGUAGES

Spanish (native) – English (advanced, TOEFLiBT 102/120) – French (advanced, French Baccalaureate) – Italian (Intermediate, B1-B2 MCER)

PROFESSIONAL EXPERIENCE

Internship – Industrial A-frame Automation, Grüum, Manchester, *UK* | January 2026 – Now

In charge of designing, building, circuiting and programming an industrial A-frame as part of my master's dissertation. Keywords: WAGO PLC, HMI design and programming, project management, industrial networks, industrial automation.

- Designed, built and programmed an industrial A-frame automation solution from concept to validation including design, component selection and testing.
- Programmed WAGO PLC and designed HMI, establishing communication between PLC and HMI for full system control.
- Generated technical documentation for maintenance and operational handover, ensuring clarity for technical and non-technical staff.
- Delivered a validated POC (proof of concept) meeting all expectations and company requirements.

Research Assistant AI and Cyber-physical security, Manchester Metropolitan University, Manchester, *UK* | December 2025 – Now

Research assistant focused on creating an industry-like control system of a microgrid to build a test environment for cyberattacks. Keywords: teamwork, industrial networks, SCADA, PLC, HIL, industrial automation.

- Programmed RTAC SEL 3555 PLC using IEC-61131-3 standard languages in AcSelerator.
- Built a SCADA system with NodeRed for real-time data visualization and control signals over Modbus TCP.
- Configured OPAL RT (microgrid simulator) for bidirectional communication using DNP3 and IEC-61850 MMS, creating a HIL (hardware in the loop) system to simulate cyberattacks on industrial control systems.
- Analyzed network packets for delay measurement and data collection to support AI-based security algorithms.

Project Manager IoT, Alhona, Bilbao, *Spain* | September 2023 – June 2025

Industry 4.0 Project Manager, focused on integrating devices and machines into IoT platforms for data analysis and rule implementation. Keywords: project management, IoT and digitalization, data analysis, critical thinking, continuous learning, leadership, industrial automation.

- Promoted from Junior Project Manager to IoT Project Manager.
- Collaborated and managed end-to-end IoT integration: device identification, signal taxonomy definition, and deployment into data acquisition platforms.
- Conducted data analysis and deployed rules to enable predictive insights.
- Authored technical and non-technical deliverables for knowledge transfer with clients, bridging communication between engineering teams and stakeholders.
- Collaborated in analyzing and testing a new IoT platform.
- Visited international client sites and collaborated with on-site teams.

Internship, CEIT, San Sebastian, *Spain* | September 2022 – December 2022

*In charge of programming the **TM5 Omron** robot. **Dissertation grade: 10/10 + Honours.** Keywords: robot programming, problem-solving, computer vision, adaptability, TMFlow.*

- Programmed TM5 Omron robot using TMFlow to develop two interactive applications.
- Implemented computer vision integration to enable the robotic arm to play tic-tac-toe against a user with unbeatable logic.
- Developed a portrait sketching application, controlling the robotic arm with an integrated camera to draw user portraits.
- Delivered both projects independently, demonstrating problem-solving, adaptability, and rapid learning of new programming environments.

EDUCATION

MSc Robotics and Automation – Manchester Metropolitan University, Manchester, *UK* | September 2025 - September 2026

Business Introduction Program – Tecnun University of Navarra, Bilbao, *Spain* | September 2023- September 2024

Program combining practical in-company training with online academic education, providing exposure to corporate environments and business management fundamentals.

Automation and Control Engineering – Polytechnic University of Milan, Milan, *Italy* | February 2023 - July 2023

International exchange program.

Industrial Electronic Engineering – Tecnun University Navarra, San Sebastian, *Spain* | September 2019 - August 2023

SKILLS

TIA Portal, CODESYS, AcSELerator, Siemens NX, Fusion, C++, C, Java, Simulink, NodeRed, Insights Hub, Adobe Programs, Microsoft Office Tools.